

Impossible Open Spaces: Exploring the Effects of Occlusion on the Noticeability of Self-Overlapping Virtual Environments

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ABSTRACT

Impossible spaces uses self-overlapping virtual architecture to increase the area of virtual environments that can be explored by walking. This paper details a preliminary study using psychophysical methods to explore if users' ability to detect overlapping virtual spaces changes when the walls of overlapping rooms and the paths connecting them are removed. The results indicate that impossible spaces can be deployed, even when no walls are presented between adjacent rooms. However, the absence of walls appears to make it easier for users to detect impossible spaces.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Walking is the default mode of locomotion for most able-bodied individuals. However, in virtual reality (VR) the ability to freely walk is constrained by the size of the physical space. This is an obstacle because virtual environments (VEs) can be virtually infinite in size. Suma et al. [3] demonstrated that this problem can be alleviated using so-called *impossible spaces*. Impossible spaces use self-overlapping virtual architecture to compress interior VEs into smaller physical tracking spaces. Notably, the compression can be performed without the user noticing it as long as the amount of overlap does not exceed the users' detection threshold. Based on psychophysical methods, Suma et al. [3] estimated how much two adjacent virtual rooms can overlap before users are able to reliably detect the manipulation. The results indicated that relatively small adjacent rooms with fixed sizes (3.7m × 7.3m) can overlap by up to 56%, and larger rooms that expanded to fill the tracking space (9.1m × 9.1m) can overlap by up to 31%. More recent work has shown that the shape of the corridor connecting self-overlapping virtual architecture can also affect users' perception [4], and it may be possible to impair users' ability detecting the manipulation by introducing distractions in the corridors connecting overlapping rooms [1]. That is, rooms could overlap by up to 68% when users were distracted, compared to 40% when no distractions were presented. Other work by Suma et al. [2] showed that visual change blindness can be used to conceal manipulations of the virtual architecture. Particularly, they showed that doors and hallways can be moved behind users' backs and thereby permit virtual spaces to overlap without the users' knowledge. However, research on the factors affecting users' ability to detect impossible spaces is scarce, and it has yet to be explored whether impossible spaces can be deployed when the user's view is not occluded by walls, as is likely to be the case in outdoor virtual environments. In this paper we describe a preliminary study exploring if users' ability to detect overlapping virtual spaces and estimate distances changes when the walls of adjacent spaces and the paths connecting them are removed.

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2 METHODS AND MATERIALS

To explore how visual occlusion affects the noticeability of impossible spaces and users distance judgments, we performed a within-subjects comparing three conditions: *Total occlusion*: The two adjacent virtual rooms and the connecting corridor had opaque walls. *Open corridor*: Both rooms were surrounded by walls, but the user was able to see the surrounding VE when walking in the corridor. *Open rooms*: The corridor was surrounded by walls, but the user could look from one room into the other. To allow the user to see adjacent virtual spaces, the walls were replaced by fences that did not obstruct the users' view. Figure 1 shows the three conditions. Seven participants took part in the study (1 female, 6 males, mean age = 25.5 years, SD=2.1).

2.1 Virtual Environments and Equipment

For each condition, the participants were exposed to six outdoor VEs, comprising stone ruins surrounded by forest. All six included two adjacent rooms connected by a corridor, but they varied in terms of how much the two rooms overlapped: 0% (no overlap), 15%, 30%, 45%, 60%, and 75%. Each overlap level was repeated 5 times, yielding a total 90 trials (3 conditions × 6 overlap levels × 5 repetitions). The VEs were presented using an Oculus Rift S and noise-cancelling headphones. The tracking space was approximately 3m × 5m, and a single Oculus controller was used for interaction.

2.2 Procedure and Measures

First the participants filled a questionnaire for demographic information and a consent form. Subsequently, they were exposed to the 90 trials, which were presented in randomized order. Each trial started in the center of one of the two adjacent virtual rooms. The participants were instructed to inspect this room before walking to the second room using the corridor. When the participants had explored the second room, they returned to the center of the first room (the centers of both rooms were highlighted using white circles on the floor). To evaluate the *noticeability* of self-overlapping virtual architecture, we relied on the psychophysical approach used in the

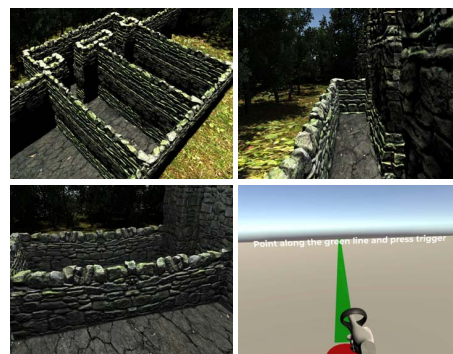


Figure 1: Overhead view of the VE with total occlusion (top left), and a user's view of the open corridor (top right), the open rooms (bottom left), and the pointing task (bottom right).

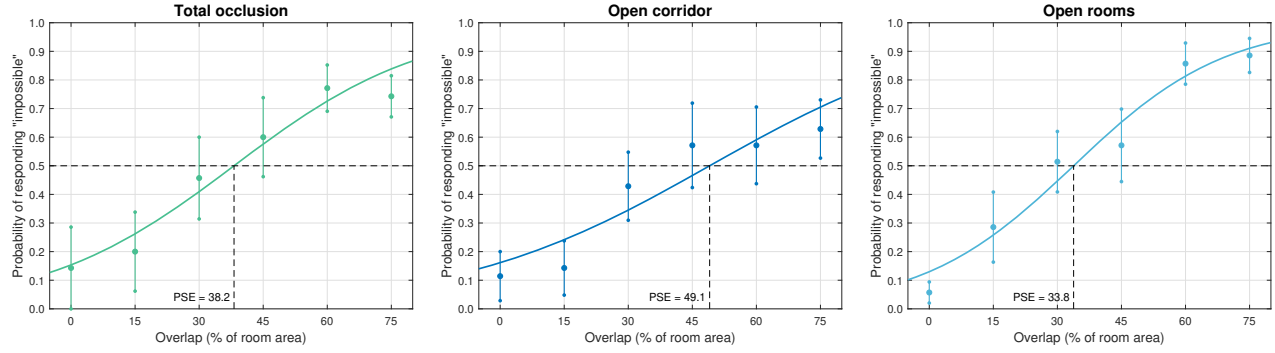


Figure 2: Detection results, standard errors, and fitted psychometric functions for the condition with total occlusion, open corridor, and open rooms. The x-axes show the percentage of overlap, and the y-axes represent the probability that participants responded that the VE was “impossible”. The horizontal dotted lines show points on the curves where the participants on average responded “impossible” half of the time (i.e., random chance), and the corresponding vertical dotted line identify the point of subjective equality (PSE).

original work by Suma et al. [3]. Particularly, we used the same six levels of overlap (0%, 15%, 30%, 45%, 60%, and 75%), and after exposure to each VE, the participants performed a two-alternative forced-choice (2AFC) task, requiring them to judge whether the VE they had just visited was “possible” or “impossible.” This judgement was made in a relatively neutral environment where the participants also performed a distance estimation task at the end of each trial (Figure 1). That is, they were asked to indicate the distance to the center of the other room by pointing to a green line extending from under feet and through the center of the other room. The study lasted approximately one hour per participant.

3 RESULTS

Given the exploratory nature of the study and the small sample size, our analysis focuses exclusively on descriptive statistics. The data related to *noticeability* were analyzed in the same manner as Suma et al. [3]. We derived the pooled probability that participants found the overlap level “impossible” for each overlap level, and fitted psychometric functions of the form $f(x) = \frac{1}{1+e^{ax+b}}$ where a and b are real numbers, for the three conditions. We defined the detection thresholds associated with each condition as the overlap level at which the participants responded at chance level (i.e., they were equally likely to respond “possible” or “impossible” on the 2AFC task). As apparent from Figure 2, the detection threshold for the condition with total occlusion was at a 38.2% overlap, for open corridor it was at a 49.1% overlap, and for open rooms it was at a 33.8% overlap. Figure 3 shows the results of the *distance estimation*

task. The participants consistently overestimate the virtual distance, and the magnitude of the overestimation appears to increase as the amount of overlap becomes larger. However, it is unclear if the amount of overestimation is affected by removal of the virtual walls.

4 CONCLUSION

First, our results indicate that impossible spaces can be deployed even if there is no opaque wall separate the overlapping rooms, and our preliminary results indicate that they may overlap by as much as 33.8%. Nevertheless, it is notable that this threshold is considerably lower than the threshold obtained when their view of the adjacent rooms was obstructed (38.2% overlap). This may indicate that the absence of walls makes the manipulation easier to detect. Surprisingly, the condition with an open corridor yielded the highest detection threshold of the three conditions (49.1% overlap). A possible explanation is that the view of the forest distracted the participants when moving from one room to another and therefore affected their judgments. This would be in line with previous work on the effects of distractions [1]. However, considering the small sample size, future work is needed to corroborate these findings. We omitted a condition where both rooms and corridors were open, to reduce the duration of the study and limit fatigue. However, in the light of the results it would be interesting for future studies to include such a condition. Finally, the study explored the effect on distance estimation, but it remains unclear if removing walls had an effect. The results suggest that the amount of the overestimation increases as the degree of overlap increases. This is in line with prior work, but the amount of the effect is less pronounced [3].

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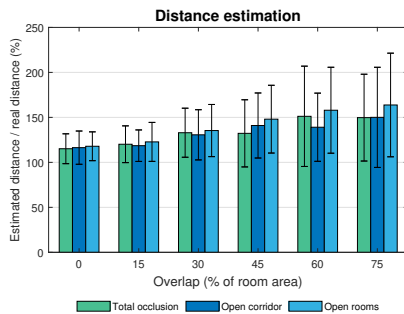


Figure 3: Mean ratio of distance estimates to real distances, expressed as percentages (percentages over 100% indicate overestimation). Error bars show 95% CIs.